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SECURITY ASSESSMENT REPORT

TryHackMe – Introductory Networking

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Overview

This room goes deeper into how computer networks work and introduces some tools for investigating network communication. It covers the OSI and TCP/IP models, how data travels through different layers, and how to use commands such as `ping`, `traceroute`, `whois`, and `dig`.

Networking is a foundation for penetration testing. Knowing how devices communicate makes it easier to understand scan results, troubleshoot connections, and investigate the services running on a target.

1. The OSI Model

The OSI model divides network communication into seven layers. Each layer has a different role, which makes it easier to understand where a particular networking problem or protocol fits.

Layer	Name	Main purpose
7	Application	Network services used by applications
6	Presentation	Data formatting, encryption, and compression
5	Session	Managing communication sessions
4	Transport	End-to-end communication using TCP or UDP
3	Network	Logical addressing and routing
2	Data Link	Communication over a local link
1	Physical	Transmitting raw bits over a physical medium

The OSI model is useful because it gives me a way to organise networking concepts instead of treating every protocol as something unrelated.

For example, IP operates at the network layer, while TCP operates at the transport layer. This distinction becomes important when looking at packet captures or interpreting the results of network tools.

2. Encapsulation

Encapsulation is the process of adding protocol information to data as it moves down the networking layers before transmission.

For example, when an application sends data using TCP over IP and Ethernet, the data is wrapped with the relevant headers at different layers. At the receiving end, those headers are processed and removed as the data moves back up the stack.

3. The TCP/IP Model

The TCP/IP model describes networking using four layers:

- **Application:** Protocols and services such as HTTP and DNS.
- **Transport:** Communication using TCP or UDP.
- **Internet:** IP addressing and routing between networks.
- **Network Interface:** Communication over the local network.

The TCP/IP model is more compact than the seven-layer OSI model and is commonly used to describe real-world networking.

Both models help explain how network communication works, but they organise the layers differently.

4. Wireshark

Wireshark is a network protocol analyser that captures and displays network traffic. It allows you to inspect packets and examine details such as source and destination addresses, protocols, ports, and packet contents when available.

This is useful when troubleshooting a connection or trying to understand what happens when a device communicates with a server.

For penetration testing, packet analysis can help explain unexpected behaviour that isn't obvious from a command's output alone.

5. Networking Tools

Ping

`ping` tests connectivity by sending ICMP Echo Requests and checking for replies.

```
ping 10.10.10.10
```

Replies indicate that the destination responded to the probes. No reply doesn't necessarily mean the host is offline, since ICMP traffic may be filtered.

Traceroute

`traceroute` helps identify the intermediate network hops between your machine and a destination.

```
traceroute example.com
```

It is useful for investigating network paths and identifying where responses stop appearing. Some routers don't respond to traceroute probes, so missing hops don't automatically indicate a connection failure.

WHOIS

`whois` retrieves available registration information about a domain or IP address.

```
whois example.com
```

Depending on the registry and the domain, results may include registrar details, registration dates, nameservers, and contact information.

This can be useful during reconnaissance when researching a domain or understanding its registration details.

Dig

`dig` is a tool for querying DNS and inspecting DNS records.

```
dig example.com
```

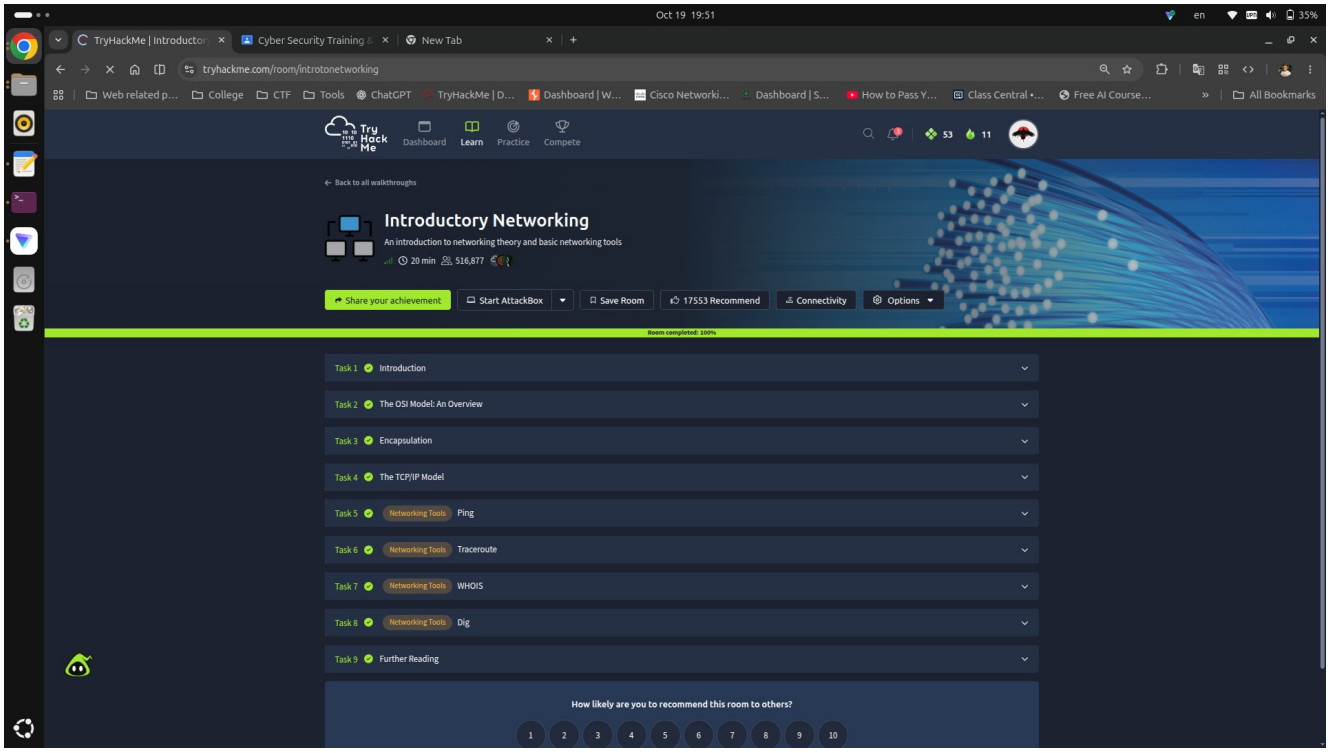
It can return information such as the resolved IP address, the DNS server that answered, and the record's time to live (TTL).

DNS translates domain names into information that network clients can use to locate services. Understanding how to query DNS is useful when investigating domains and troubleshooting name-resolution problems.

Conclusion

This room helped connect networking theory with tools that can be used to investigate real network behaviour. The OSI model and encapsulation explain what's happening behind the scenes, while tools such as Wireshark, `traceroute`, and `dig` provide ways to examine different parts of that process.

These concepts are particularly relevant to penetration testing because reconnaissance often starts with understanding a target's network, identifying services, and gathering information before moving on to further testing.


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